Fatemeh Barati

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Education

Kharazmi University **2023 - 2025**
Master's in Computer Science (Decision and Knowledge Sciences), GPA: 18.04/20 *Karaj, Iran*
Courses: Machine Learning (18/20), Deep Learning (20/20), Data Mining (19.5/20), Stochastic Processes (19.5/20), Advanced Algorithm (18/20), Seminar (20/20)

Thesis: **Explainable Artificial Intelligence (XAI) for Convolutional Neural Networks (CNNs)** — 19.75/20 (Excellent).

Introduced Contrastive Gradient-Flow Interpretation (CGFI), a novel XAI method combining gradient-flow analysis with contrastive learning to generate faithful and discriminative explanatory maps for CNN predictions.

- Thesis Advisor: Prof. Mohammad Soltanian

Alzahra University **2019 - 2023**
Bachelor's in Electrical Engineering, GPA: 16.58/20 *Tehran, Iran*
Bachelor's Project: Finger Detection and Counting using AI

Implemented a computer vision application using OpenCV for real-time finger counting through hand region segmentation, contour feature extraction, and live display.

- Project Advisor: Prof. Amire Seyedfaraji

Technische Universität Berlin **Oct 2021 - Dec 2021**
Blue Engineering *Berlin, Germany (Remote)*
An international and innovative workshop-style course (6 ECTS) focusing on ecological and social responsibility in engineering practice.

Research Interests

Human-Centered AI and Human-AI Collaboration

Explainable, Fair, and Trustworthy AI (XAI)

Interdisciplinary Applications of ML/DL (Behavioral Studies, HCI, AI for Education, AI for Society)

Honours and Awards

MenaML Winter School 2026 - Full Sponsorship Award **Jan 2026**
King Abdullah University of Science and Technology (KAUST), Saudi Arabia
- Selected as 1 of 300 scholars from 2,300 applicants (13% acceptance) for intensive AI/ML training at KAUST by experts from Google DeepMind, QCRI, and InstaDeep; sponsored by Google, Google DeepMind, Apple, and NVIDIA, etc.

Publications

- **F. Barati**, M. Soltanian, K. Borna, and L. Longo, "Contrastive Gradient-Flow Interpretation (CGFI): A Composable Operator for Class-Discriminative Explanation of Convolutional Neural Networks," *Machine Learning and Knowledge Extraction*, vol. 8, no. 10, p. 304, 2026. [DOI]
- **F. Barati**, "Human-Centered Explainable AI: An Overview of Current Approaches and Challenges," in *Proceedings of the Second International Conference on Computer, Electrical, Mechanical, and Engineering Sciences*, May 2025. [Link]

Teaching Experience

Kharazmi University **Sep 2024 - Feb 2025**

Head Teaching Assistant of Machine Learning *Karaj, Iran*

- Led TA team for Prof. Keivan Borna's Master's-level course, developing curriculum materials and exercises, proposing exam questions, grading assignments and final projects, and providing individualized student support.

Kharazmi University **Sep 2024 - Feb 2025**

Head Teaching Assistant of Deep Learning *Karaj, Iran*

- Led TA team for Prof. Mohammad Soltanian's English-taught course for international Master's students, delivering tutorial sessions and designing exercises to reinforce theoretical concepts through practical implementation.

Kharazmi University **Sep 2024 - Feb 2025**

Head Teaching Assistant of Advanced Data Mining *Karaj, Iran*

- Led TA team for Prof. Bardia Panahbehagh's English-taught course for international Master's students, developing comprehensive curriculum materials with hands-on coding exercises, conducting tutorial and problem-solving sessions, and providing individualized student support.
- Administered and graded final live-coding exams, and assessed assignments and final projects.

Work Experience

Mobile Communications of Iran (Hamrahe Aval) **Jul 2025 - Nov 2025**

Technical Product Management Intern *Tehran, Iran*

- Redesigned the end-to-end journey of an on-site SIM service (about **12,000 monthly active customers**), where a technician visits to fix, transfer, or issue a SIM: landing page, sign-up, user portal, visit, and feedback.
- Led user research and designed surveys for the employees who run the service.
- Wrote Product Requirement Documents (PRDs) for sign-up workflows with multiple scenarios and streamlined service ordering.
- Worked with the lead Technical Product Manager, stakeholders, developers, and operators to prioritize the roadmap and drive feature implementation.
- Trained in Microsoft Dynamics 365 and Jira.

British Airways **Jun 2024**

Junior Data Scientist *London, United Kingdom (Remote)*

- Completed a job simulation focusing on how data science is a critical component of British Airways success.
- Scraped and analyzed customer review data to uncover findings.
- Built a predictive model to understand factors that influence buying behavior.

MECO (Mapna Electric & Control, Engineering & Manufacturing) **Aug 2022 - Sep 2022**

AI Intern *Karaj, Iran*

- Preprocessed turbine's data for machine learning models.
- Trained a multiclass classification model to analyze turbine's data.
- Implemented Decision Tree, Random Forest, SVM, and KNN algorithms.

Alzahra University **Jun 2021 - Jun 2022**

Head of Research Committee at Electrical Engineering Student's Scientific Association *Tehran, Iran*

- Supervised a team consisting of tutors, translators, and content producers.
- Created educational videos and social media posts to enhance students' knowledge.
- Translated English articles in the Electrical Engineering field into Persian.
- Made Persian subtitles for English videos about key concepts in Electrical Engineering and modern technologies.

Volunteering

The 17th International Conference of Iranian Operations Research Society **Jul 2024**

Executive Staff *Tehran, Iran*

- Served on the executive team for the conference hosted by Kharazmi University of Tehran, assisting with preparation of participant badges, organizing awards ceremonies, and coordinating distribution of medals and prizes for distinguished professors and best paper recipients.

Selected Projects

Heart Disease Prediction **Jun 2024 - Jul 2024**

Kharazmi University | Advanced Data Mining Course

- Processed and analyzed a Kaggle heart disease dataset, selecting key risk factors for prediction.
- Applied and compared various machine learning classifiers, including Decision Trees, Support Vector Machines, and Neural Networks, to assess their performance.
- Extracted insights from model results to identify individuals at high risk of heart disease.

Image Classification using Transfer Learning **Jul 2024**

Kharazmi University | Deep Learning Course

- Implemented an image classification project using the Transfer Learning technique to enhance model performance by leveraging pre-trained models such as EfficientNet and ResNet.
- Integrated data augmentation strategies to enhance the model's ability to generalize and improve accuracy.
- Applied multiple regularization techniques to reduce overfitting and improve model stability.

AI-Driven Fashion Recommender System **Aug 2023 - Oct 2023**

Institute for Research in Fundamental Sciences (IPM) | Artificial Intelligence Summer Camp

- Contributed to the development of an AI-driven fashion recommender system within a team. My role involved training classification models to categorize clothing by gender and season.

Finger Detection and Counting using AI **Jul 2023 - Aug 2023**

Alzahra University | Bachelor's Project

- Developed a real-time finger detection and counting system using OpenCV by designing a pipeline to locate the hand region, extract contour features, and display the finger count live.

Turbine's Data Classification using AI **Aug 2022 - Sep 2022**

MECO (Mapna Electric & Control, Engineering & Manufacturing) | Internship Project

- Implemented multiple classification models (Random Forest, KNN, Decision Tree, SVM) on a dataset of over 20,000 rows and 45 features.
- Achieved 99% accuracy using Random Forest, significantly outperforming other classifiers.

Technical Skills

Fields: Machine Learning, Deep Learning, Computer Vision, Explainable AI (XAI)
Programming Languages: Python, C, C++
Python Libraries: TensorFlow, Keras, PyTorch, OpenCV, Scikit-learn, Pandas, Numpy, Plotly, Matplotlib
Front-end: HTML, CSS, JavaScript, Bootstrap, jQuery

Selected Certifications

Behavioral Economics
MCI Academy | Apr 2026

AI Ethics
DataCamp | Feb 2025

Explainable AI in Python
DataCamp | Feb 2025

Explainable Artificial Intelligence (XAI) Concepts
DataCamp | Nov 2024

Intermediate SQL
DataCamp | Sep 2024

How Diffusion Models Work
DeepLearning.AI | Aug 2024

Image Processing in Python
DataCamp | Nov 2023

Artificial Intelligence Summer Camp
Institute for Research in Fundamental Sciences (IPM) | Oct 2023

Introduction to Git and Github
Coursera | Sep 2023

Python for Computer Vision with OpenCV and Deep Learning
Udemy | Aug 2023

Machine Learning Specialization
Coursera | Apr 2023

Data Cleaning
Kaggle | Sep 2022

Languages

- **Persian**: Native
- **English**: IELTS Academic — Overall Band **7.0** (Listening 8.0, Reading 7.5, Writing 6.5, Speaking 6.5)
  Test Date: April 15, 2025

References

Prof. Luca Longo **Contact: luca.longo@ucc.ie**
Associate Professor at University College Cork, Ireland

Prof. Mohammad Soltanian **Contact: m.soltanian@khu.ac.ir**
Assistant Professor at Kharazmi University, Iran

Prof. Keivan Borna **Contact: borna@khu.ac.ir**
Associate Professor at Kharazmi University, Iran

Prof. Bardia Panahbehagh **Contact: panahbehagh@khu.ac.ir**
Assistant Professor at Kharazmi University, Iran